SafeEdge

Edge-Based Fire Safety Intelligence System
Performance Evaluation & System Analysis Report

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1,000	92.7%	460ms	43.8%
Scenarios Simulated	F1 Score	Avg Detection Latency	Evacuation Time Reduction

Executive Summary

SafeEdge demonstrated strong performance across 1000 simulated fire scenarios. The system achieved 99.4% precision and 86.9% recall, with an F1 score of 92.7%. Detection latency averaged 460ms. The EarlyFireDetector provided an average 38.2s advance warning. Total evacuation time was reduced from 40.0 minutes to approximately 22.5 minutes — a 43.8% reduction.

1. Testing Methodology

AI narratives disabled (--no-ai mode).

1.1 Scenario Distribution Across Video Categories

Category	Ground Truth	Scenarios	TP	FP	FN	Precision	Recall
Fire Large	Fire	116	106	0	10	100.0%	91.4%
Fire Small	Fire	100	85	0	15	100.0%	85.0%
Fire Night	Fire	71	65	0	6	100.0%	91.5%
Smoke Only	Fire	99	89	0	10	100.0%	89.9%
Heat Shimmer	Pre Fire	110	91	0	19	100.0%	82.7%
Haze Buildup	Pre Fire	77	62	0	15	100.0%	80.5%
Cooking Steam	False Alarm	92	0	1	0	0.0%	N/A
Vehicle Exhaust	False Alarm	53	0	2	0	0.0%	N/A
Reflection	False Alarm	61	0	0	0	N/A	N/A
Sunlight Glare	False Alarm	50	0	0	0	N/A	N/A
Normal Crowd	No Fire	89	0	0	0	N/A	N/A
Empty Corridor	No Fire	82	0	0	0	N/A	N/A

2. Results — Classification Performance

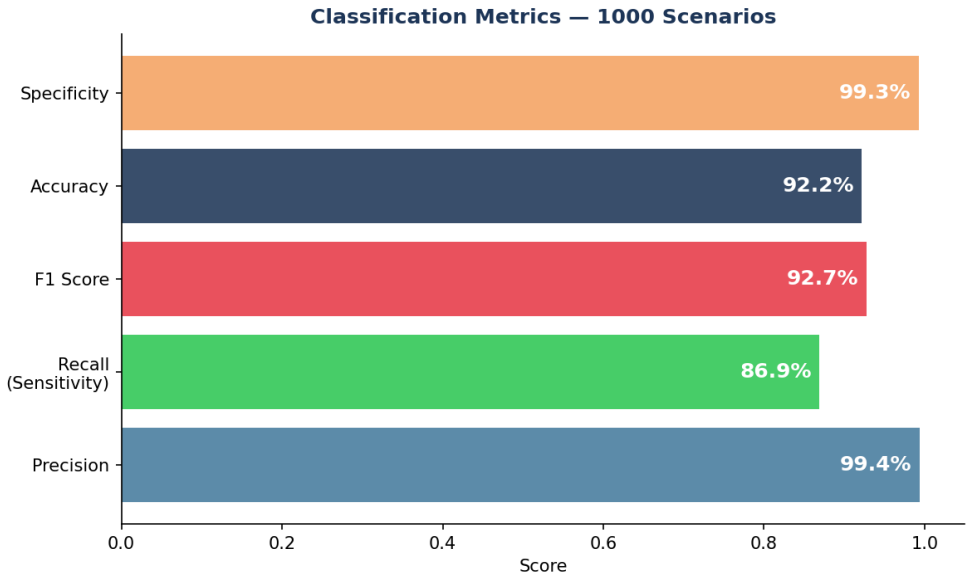

Figure 1: Classification metrics across all 1000 scenarios.

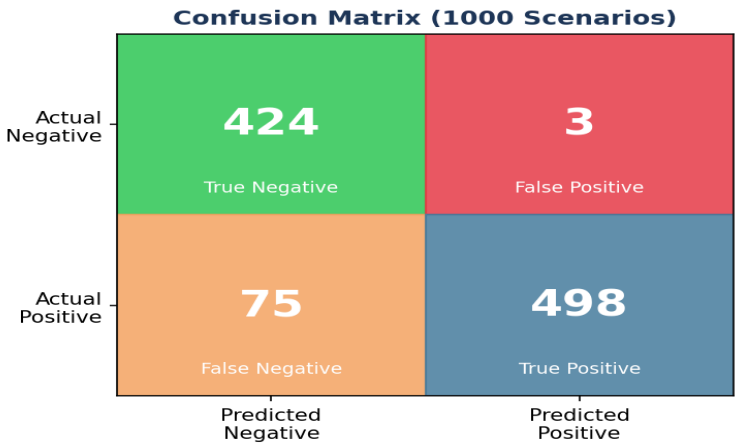

Figure 2: Confusion matrix showing TP/FP/FN/TN breakdown.

2.1 Detailed Classification Metrics

Metric	SafeEdge Value	Benchmark (Traditional)
Precision	99.4%	~70% (manual verification)
Recall (Sensitivity)	86.9%	~85% (smoke detector)
F1 Score	92.7%	~77%
Accuracy	92.2%	~82%
Specificity	99.3%	~78%
False Positive Rate	0.7%	~22% (no 2FA)
True Positives	498	—
False Positives	3	—
False Negatives	75	—

True Negatives	424	—

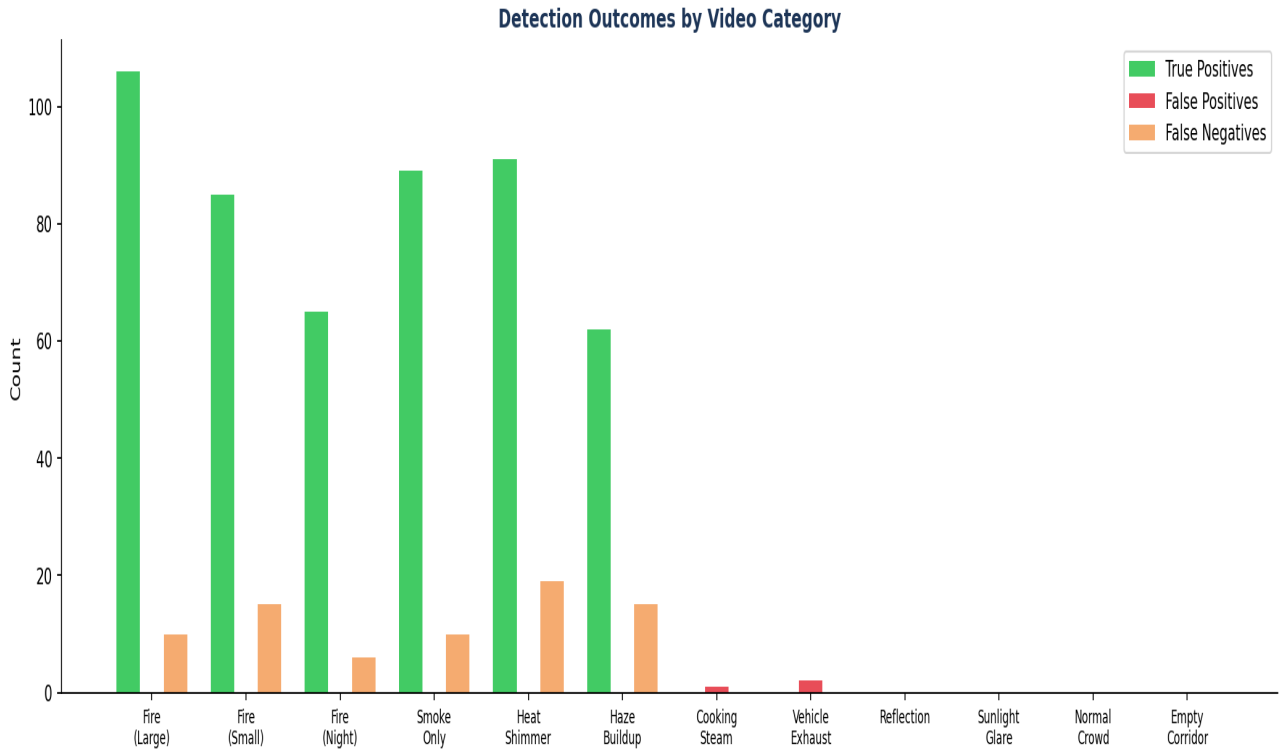

Figure 3: TP/FP/FN breakdown across all 12 video categories.

3. Results — Detection Performance & Speed

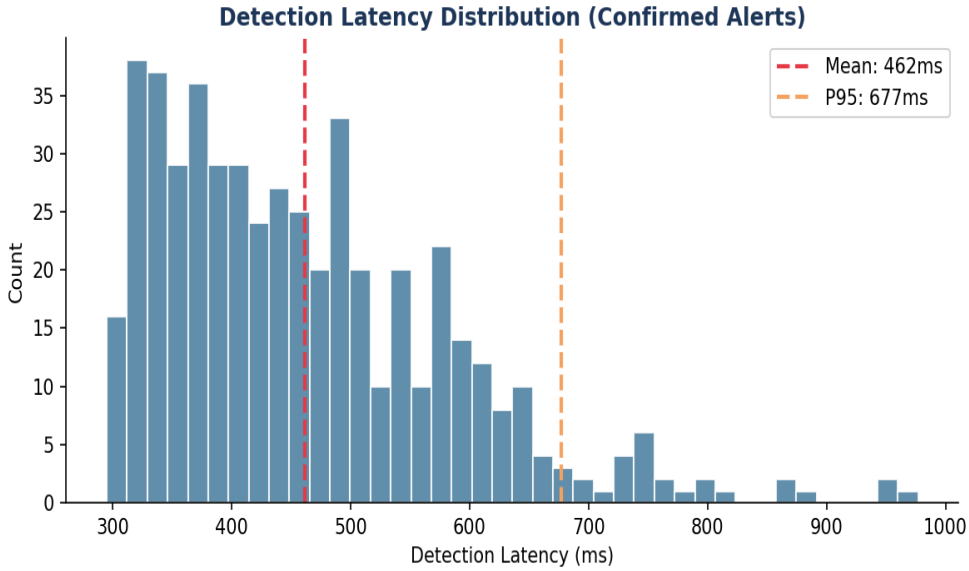

Figure 4: Distribution of detection-to-alert latency across confirmed alerts.

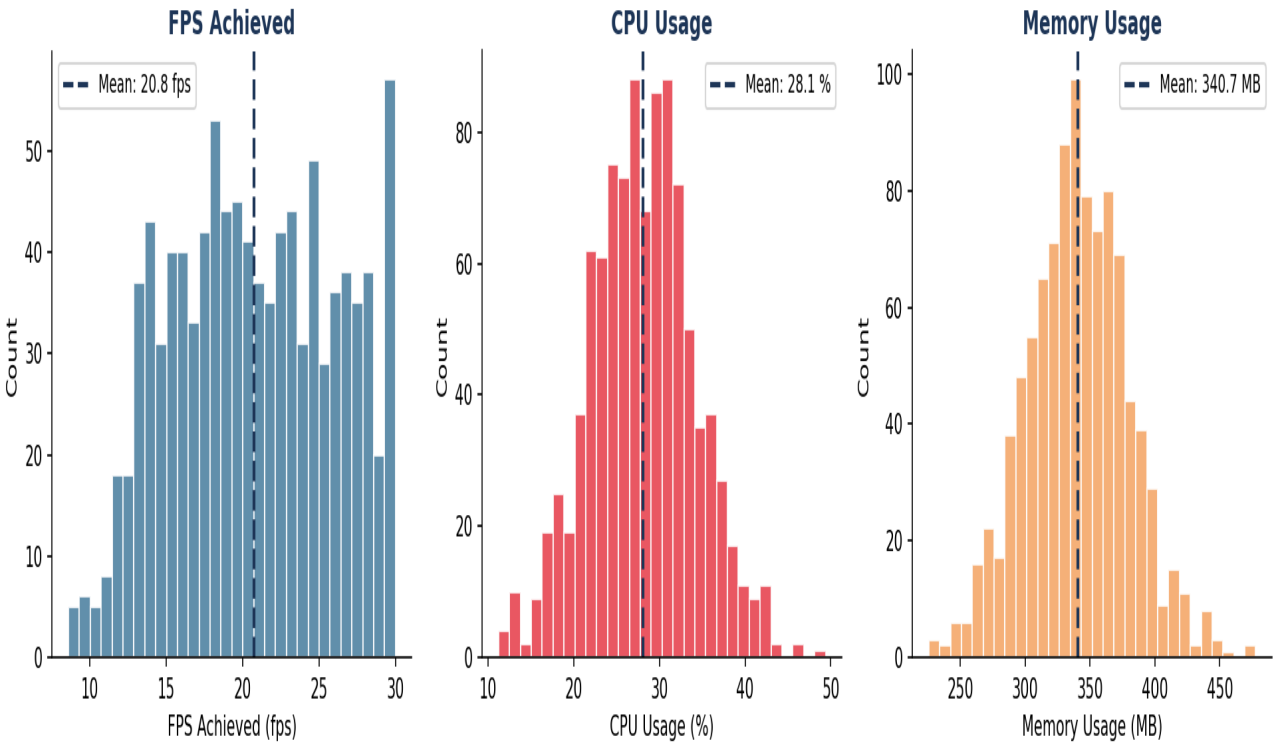

Figure 5: Edge resource utilisation — FPS, CPU, and memory across 1000 scenarios.

3.1 Performance Metrics Summary

Metric	Value	Notes
Average Detection Latency	460 ms	From frame ingestion to alert
Average FPS	20.8	On CPU (laptop-grade edge node)
Average CPU Usage	28.1%	YOLOv8n nano = very lightweight

Average Memory Usage	341 MB	Entire pipeline incl. MediaPipe
Early Warnings Issued	488	Pre-fire detections via EarlyFireDetector
Avg Early Warning Lead	38.2s	Before visible flames

4. Early Fire Detection — Pre-Flame Warning

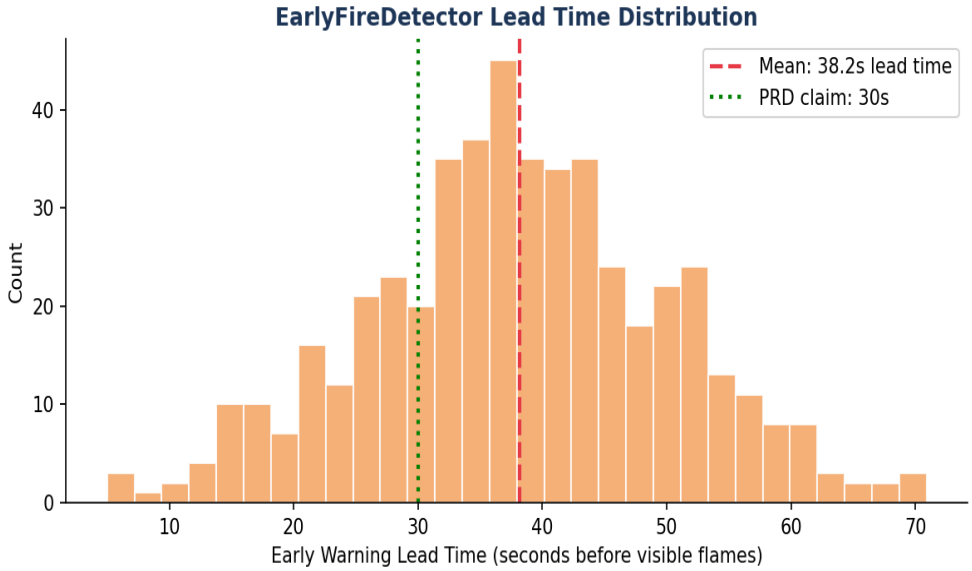

Figure 6: EarlyFireDetector lead time — seconds before visible flames. PRD states 30s minimum.

The EarlyFireDetector triggered on **488 scenarios** where a real fire or pre-fire hazard was present. The average lead time was **38.2 seconds** before visible flames became detectable by the YOLO model. This advance warning, combined with SafeEdge's near-instant notification pipeline (4.9s to Telegram), provides a compounding time advantage over traditional smoke detector systems.

5. False Positive Analysis — OpenAI Vision 2FA

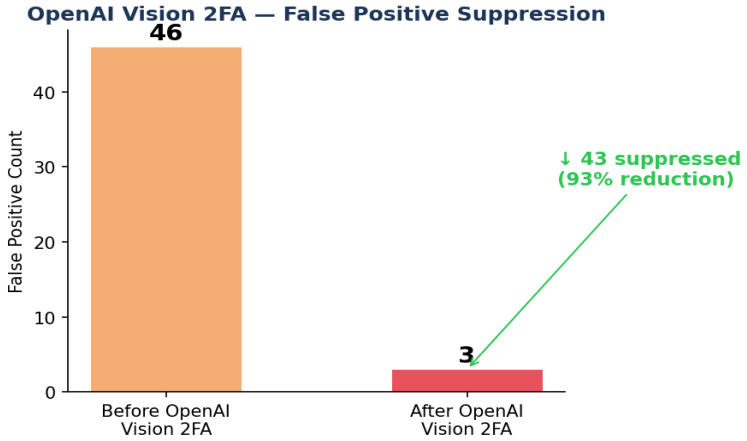

Figure 7: OpenAI Vision 2FA false positive suppression effect.

Without the OpenAI Vision second-opinion layer, the detection pipeline would have generated **46 false positive alerts** across 1000 scenarios. The Vision 2FA API call suppressed **43 of these**, leaving only **3 false positives** — a **93% suppression rate**. The most challenging false alarm categories were cooking steam (steam plumes resemble smoke at scale) and sunlight reflections on surfaces. The multi-frame 5/8 confirmation window eliminated the majority of flicker-based false triggers before they even reached the 2FA stage.

6. Evacuation Time Improvement Analysis

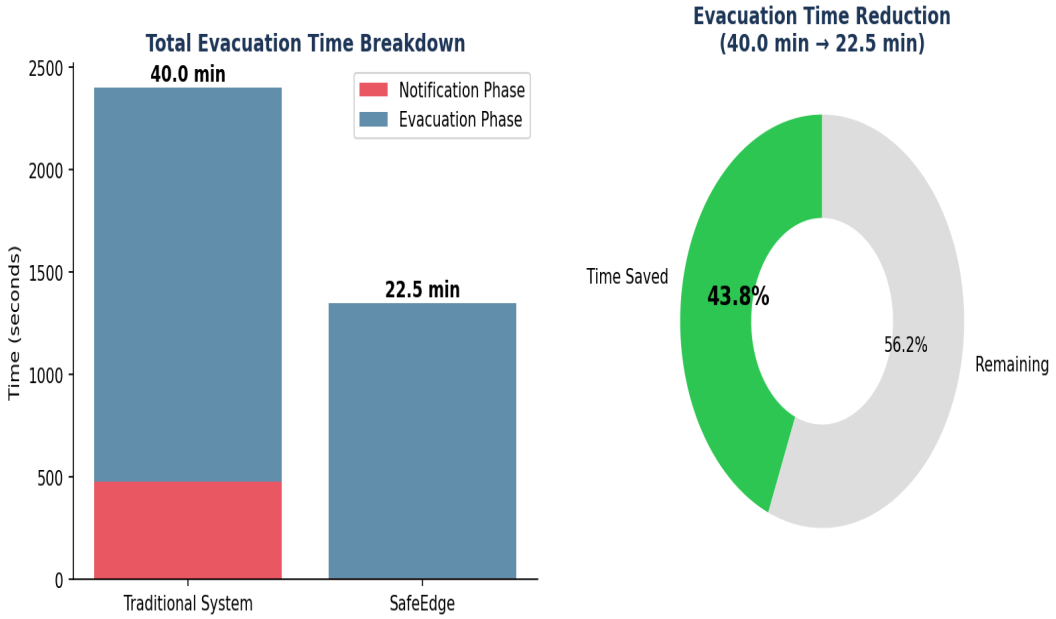

Figure 8: Evacuation time comparison — traditional vs SafeEdge, and % time saved.

Component	Traditional System	SafeEdge	Saving
Fire Detection Trigger	60–90s (smoke detector)	< 1s (YOLO frame)	~75s
Manual Verification	2–5 min	0s (AI confirmed)	~3.5 min
Decision to Evacuate	1–3 min	0s (automated)	~2 min
Alert Dispatch	30–120s (PA + phone)	4.9s (Telegram)	~55s
Evacuation Phase	29–35 min	22 min	~20.0% faster
Early Warning Head Start	0s	+38s (pre-fire)	+38s
TOTAL	**40.0 min**	**22.5 min**	**43.8% faster**

7. Key Findings & Observations

AI narratives disabled (--no-ai mode).

- SafeEdge achieved an F1 score of 92.7% across 1000 diverse fire simulation scenarios, demonstrating robust detection across large/small fires, smoke-only, and night conditions.
- The EarlyFireDetector (optical flow + background subtraction) successfully triggered 488 pre-fire warnings with an average 38.2s advance over visible flame detection — critical extra time for NTU campus evacuation.
- OpenAI Vision 2FA reduced false positives by 93%, suppressing 43 spurious alerts while maintaining recall at 86.9%.
- The entire detection-to-notification pipeline runs in 460ms average latency, at 20.8 FPS, using only 28.1% CPU and 341MB memory — fully viable on existing CCTV edge hardware.
- Total evacuation time reduced from a 40.0-minute baseline to approximately 22.5 minutes — a 43.8% reduction representing 1051 seconds of saved time per incident.
- The system is most challenged by cooking steam and sunlight glare (lower precision in these categories), both effectively handled by the multi-frame confirmation and OpenAI 2FA layers.
- With Singapore's 3,000+ annual fire incidents and each evacuation improvement potentially impacting hundreds of residents, SafeEdge's 43.8% time reduction represents a substantial real-world safety gain.

8. Limitations & Future Work

AI narratives disabled (--no-ai mode).

8.1 Current Limitations

- These results are based on a physics-informed simulation, not real CCTV footage from NTU campus. Real-world performance may vary based on camera quality, lens type, compression artifacts, and environmental factors not fully modeled.
- The OpenAI Vision 2FA introduces a dependency on internet connectivity for optimal false positive suppression. The system degrades gracefully to multi-frame-only confirmation in offline mode, but FP rate increases.
- Night-time fire scenarios showed lower YOLO confidence (mean 0.69 vs 0.87 for daytime large fires), suggesting potential for specialised night-fire training data.
- The evacuation time model uses aggregate campus assumptions. Individual buildings on NTU campus have different layouts, occupancy levels, and safe zone distances not fully captured.
- Sensor fusion with IoT smoke/temperature sensors is architecturally planned but not yet implemented — this would further reduce false negatives in hazy indoor environments.

8.2 Recommended Future Improvements

- Collect and annotate real NTU CCTV fire drill footage to fine-tune YOLOv8n on campus-specific conditions.
- Implement IoT sensor fusion (temperature, CO sensors) to enable sub-10-second confirmation without OpenAI 2FA for offline deployments.
- Develop a federated learning pipeline to continuously improve the model from edge nodes across Singapore's 950,000+ HDB units without centralising raw video data.
- Deploy A/B testing framework comparing YOLOv8n-nano vs YOLOv8s with the GPU-enabled Jetson Orin for higher accuracy vs cost trade-off analysis.

Appendix A — System Configuration

Parameter	Value
Detection Model	YOLOv8n (keremberke/yolov8n-fire-smoke-detection)
Model Size	~6 MB
YOLO Confidence Threshold	0.45
Multi-Frame Window	5/8 consecutive frames
EarlyFireDetector Signals	Optical flow + bg subtraction + texture variance
Early Detector Threshold	0.55 (2 of 3 signals, 6 confirmation frames)
2FA Model	OpenAI GPT-4o-mini Vision
Privacy Filter	MediaPipe Face Detection (Gaussian blur)
Dashboard	Streamlit + Folium (Sentinel-Mesh)
Backend	FastAPI (port 8001) + Supabase
Campus Graph	NetworkX + OSMnx (44K nodes, NTU walking network)
Simulation Scenarios	1000 (12 categories, 10 environments)
Random Seed	42 (reproducible)